

Why Bengaluru Needs Fewer Cars, Not Just Better Cars

Evaluating demand-side mobility solutions for a more efficient and sustainable city.

Abstract

Bengaluru has become one of the most congested cities in the world, consistently ranking high in traffic delays and congestion. A major but often overlooked contributor is the widespread use of private cars carrying only one person for routine commuting. When lakhs of single - occupancy cars enter the road network every day, limited urban road space is used inefficiently, resulting in slower traffic movement, longer travel times, parking pressure, and rising congestion across the city.

It not only creates congestion but also creates higher per-passenger fuel use and emissions when compared to public transport or shared mobility. In the recent era, usage of electric vehicles over petrol vehicles has gained significant attention as a progressive solution. While electric vehicles may reduce tailpipe emissions, they do not remove congestion, reduce parking demand, or solve road-space inefficiency. There are other factors like environment and supply-chain such as battery material extraction, energy-source dependency, and lifecycle impacts that needs to be considered.

This suggests that urban mobility problems cannot be solved simply by replacing petrol vehicles with EVs without any change in travel behaviour. Though the type of vehicles changes, the number remains the same and hence it leads to continued structural congestion.

This study uses publicly available mobility and traffic data to evaluate how significantly single-occupancy private cars affect Bengaluru's traffic and transport system. The important objective is to quantify the city-level impact of avoidable single-occupancy private car commuting and increase public awareness of how individual travel choices collectively shape congestion levels.

Using traffic and mobility data for the period 2020 to 2025, the study analyses congestion trends, travel efficiency, and estimated transport emissions. Forecasting techniques were used to project future congestion levels under a business-as-usual scenario. Elasticity analysis was then applied to estimate how congestion responds to changes in congestion-relevant private car trips. Scenario modelling was used to assess the potential impact of trip reduction, electric vehicle transition, and combined policy interventions. Bengaluru was also benchmarked against Zurich, Singapore, and Amsterdam to compare its mobility performance with internationally recognized transport systems.

The findings suggest that Bengaluru's congestion problem is strongly linked to growing private car demand and low vehicle occupancy. Even modest reductions in unnecessary car trips could produce measurable improvements in congestion and emissions. The paper concludes that Bengaluru's long-term mobility strategy should place greater emphasis on reducing avoidable car dependence through demand management, public transport integration, and flexible work patterns rather than relying solely on cleaner vehicles or additional road infrastructure.

Introduction

Bengaluru is widely recognized as India's technology capital and one of the country's most economically dynamic cities. Over the last two decades, rapid growth in employment, population, and urban expansion has transformed the city into a major centre of innovation and investment. However, this growth has also brought severe transport pressures. Congested roads, unpredictable travel times, declining traffic speeds, and increasing commuter stress have become part of everyday life for residents.

Traffic congestion in Bengaluru is no longer an occasional inconvenience; it has become a structural urban challenge. Despite investments in metro rail, flyovers, road widening, and junction improvements, congestion levels have remained high and, in some years, have worsened further. This

suggests that infrastructure expansion alone may not be sufficient to keep pace with rising travel demand.

One of the less discussed causes of congestion is the large number of private cars carrying only one person during peak commuting hours. When thousands of single-occupant vehicles compete for limited road space each day, the transport network becomes increasingly inefficient. A road lane used by forty people travelling in separate cars occupies significantly more space than the same forty people travelling by bus, metro, or shared transport. The issue, therefore, is not only the number of vehicles, but the low occupancy of those vehicles.

Recent debates on urban transport often place strong emphasis on electric vehicles. While EV adoption is important from an emissions perspective, it does not directly reduce congestion, parking demand, or road-space scarcity. A traffic jam made up of electric cars remains a traffic jam. For a city like Bengaluru, the mobility challenge must therefore be viewed through both environmental and spatial efficiency lenses.

This paper argues that Bengaluru needs fewer unnecessary private car trips, not simply cleaner cars. It investigates how strongly congestion is linked to private car demand and whether demand-side measures can produce meaningful benefits. Using publicly available data, forecasting methods, elasticity analysis, scenario modelling, and international benchmarking, the study evaluates practical pathways to improve mobility outcomes in Bengaluru.

The central question addressed is straightforward: can Bengaluru move more people with fewer cars?

Literature Review

Urban congestion has been studied extensively across economics, transport planning, and environmental policy. Most research agrees that congestion imposes significant costs on cities through lost productivity, longer travel times, fuel wastage, stress, and increased emissions. As urban populations grow, cities that rely heavily on private vehicles often struggle to maintain efficient movement without major intervention.

Traditional responses to congestion have focused on increasing road capacity through flyovers, wider roads, and new corridors. However, transport literature has repeatedly shown that additional road space often produces only temporary relief. As travel becomes easier in the short term, more people choose to drive, leading to renewed congestion over time. This phenomenon, commonly described as induced demand, has reshaped how many planners think about road expansion.

At the same time, environmental policy has increasingly promoted electric vehicles as a pathway to decarbonising transport. EVs can reduce local air pollution and lower tailpipe emissions, particularly when supported by cleaner electricity systems. Yet several studies note that electrification does not solve many of the broader urban costs associated with car dependence. Congestion, parking demand, land consumption, crash exposure, and long travel times remain even when vehicles become cleaner.

For this reason, more recent research has shifted toward demand-side mobility strategies. These include congestion pricing, parking reform, public transport investment, shared mobility, cycling infrastructure, walkable neighbourhood design, and remote or hybrid work arrangements. The objective is not simply to change the type of vehicle used, but to reduce unnecessary car trips and increase the efficiency of urban movement.

International city experiences offer useful lessons. Singapore is frequently cited for combining vehicle ownership restraint with congestion pricing and high-quality public transport. Zurich is known for reliable and integrated multimodal transport that makes car ownership less necessary for many residents. Amsterdam has demonstrated how cycling infrastructure, pedestrian-friendly planning, and compact land use can significantly reduce car dependence while maintaining accessibility.

In the Indian context, congestion studies often focus on infrastructure deficits, traffic operations, or public transport capacity. While these are important, fewer studies specifically examine the role of single-occupant private car commuting as a measurable driver of congestion in Bengaluru. There is also limited local analysis comparing demand-side interventions with cleaner-vehicle pathways in an integrated framework.

This paper seeks to contribute to that discussion by combining traffic trend analysis, scenario modelling, and international benchmarking to assess whether reducing low-occupancy private car travel could provide stronger overall outcomes for Bengaluru.

Sources

This study relies on publicly available data and official planning documents to evaluate mobility conditions in Bengaluru.

- TomTom Traffic Index reports for congestion levels, global city rankings and travel times for Bengaluru and the Benchmarking cities.
- Comprehensive Mobility Plan (CMP), Bengaluru – for travel demand, transport patterns, and planning recommendations.
- Directorate of Urban Land Transport (DULT), Karnataka – for policy frameworks and urban mobility initiatives.
- Public vehicle registration statistics – for understanding growth in private vehicle ownership.

These sources were selected to provide traffic data, transport trends, and policy direction relevant to Bengaluru's congestion challenge.

Data

A 5- year traffic data is taken from 2020 to 2025 with various traffic parameters such as Congestion level, Global rank, average time taken to travel 10 Kms, Rush hours speed, average trip km. This is taken from the TomTom Traffic index reports, wherever available.

Congestion-relevant car trips per day and Estimated total private car trips per day are calibrated using car ownership growth, congestion trends, due to limited publicly available annual private-car-only trip data. Multiple trip-volume scenarios were considered, reflecting uncertainties in demand estimation under different urban traffic conditions.

Emission factor

Bengaluru was assigned a higher effective emission factor (0.21 kg CO₂e/km) to capture congestion-induced inefficiency, heterogeneous fleet age, and lower average operational fuel efficiency.

Singapore was assigned a lower fleet-average emission factor (0.16 kg CO₂e/km) due to reflecting stricter fleet controls, newer vehicles, and relatively efficient urban traffic operations.

Zurich was assigned 0.17 kg CO₂e/km to reflect a comparatively efficient passenger-vehicle fleet under stringent European emissions standards.

Amsterdam was assigned 0.18 kg CO₂e/km representing an efficient but mixed urban fleet with remaining conventional vehicle usage.

Vehicle Kilometres Travelled (VKT- Kms/year) is calculated as Estimated private car trips/day * Average trip km * 365 days.

CO₂ emissions(Kg/year) are calculated as VKT * Emission factor

Introduction - Why Private Car Dependence Is a Core Issue?

Private cars provide convenience and comfort. But when used heavily for routine commuting, they create traffic chaos.

Even if private cars become electric Congestion remains, Parking demand remains, Land use pressure remains, Delays remain. Therefore, EVs alone can't solve traffic efficiency.

Root Causes of High Car Use

Many commuters choose private cars because of:

- Comfort and privacy.
- Time control
- Weak last-mile connectivity
- Unreliable public transport
- Parking availability in offices
- Fixed office attendance culture

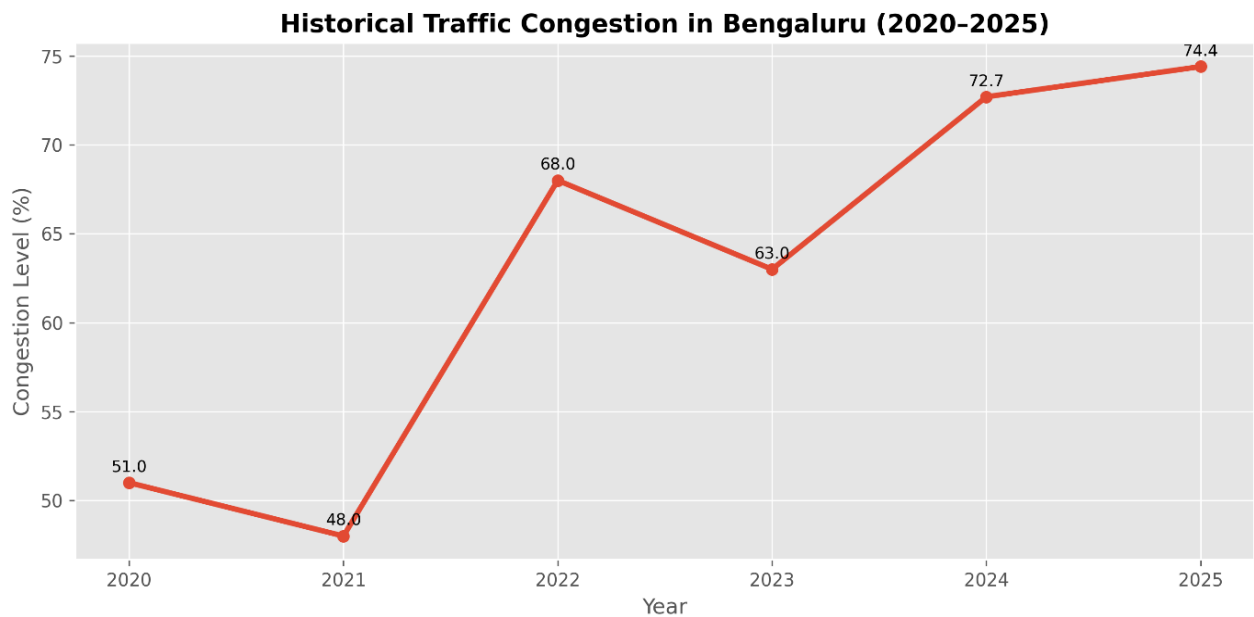
Methodology and Findings

Machine Learning model for congestion prediction

We first analysed the effect of single occupant cars on Traffic mobility focussing on Bengaluru.

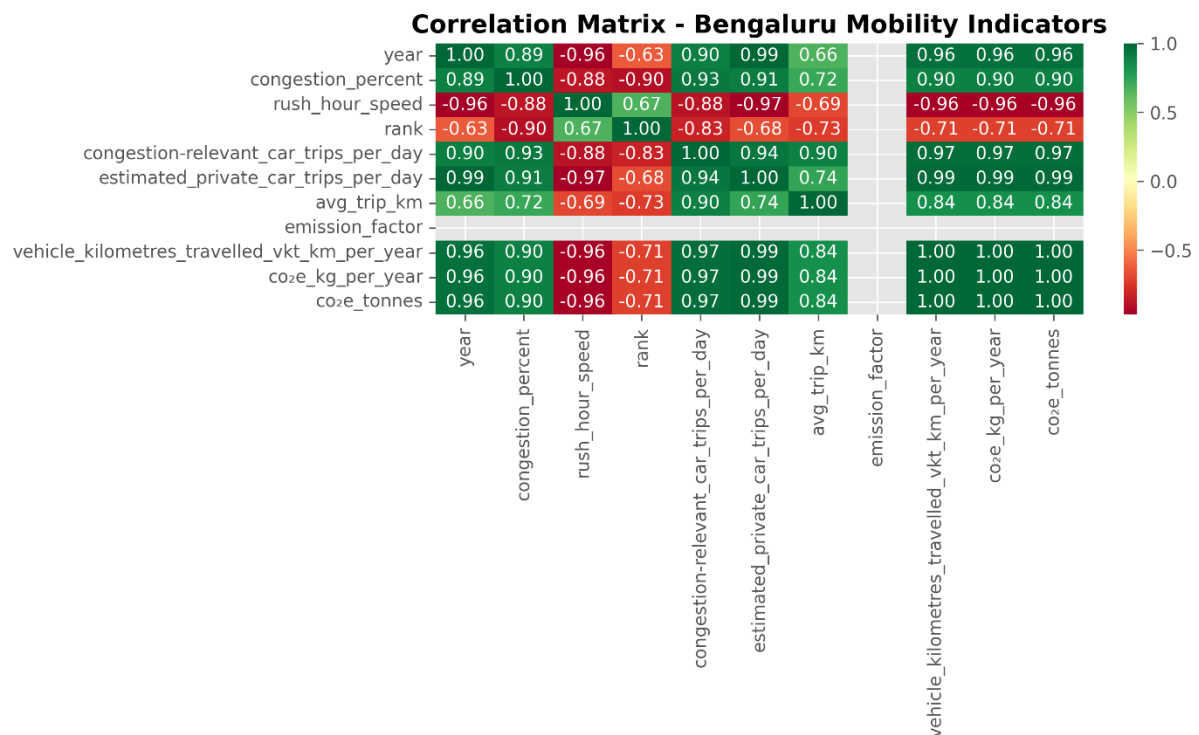
The trend analysis

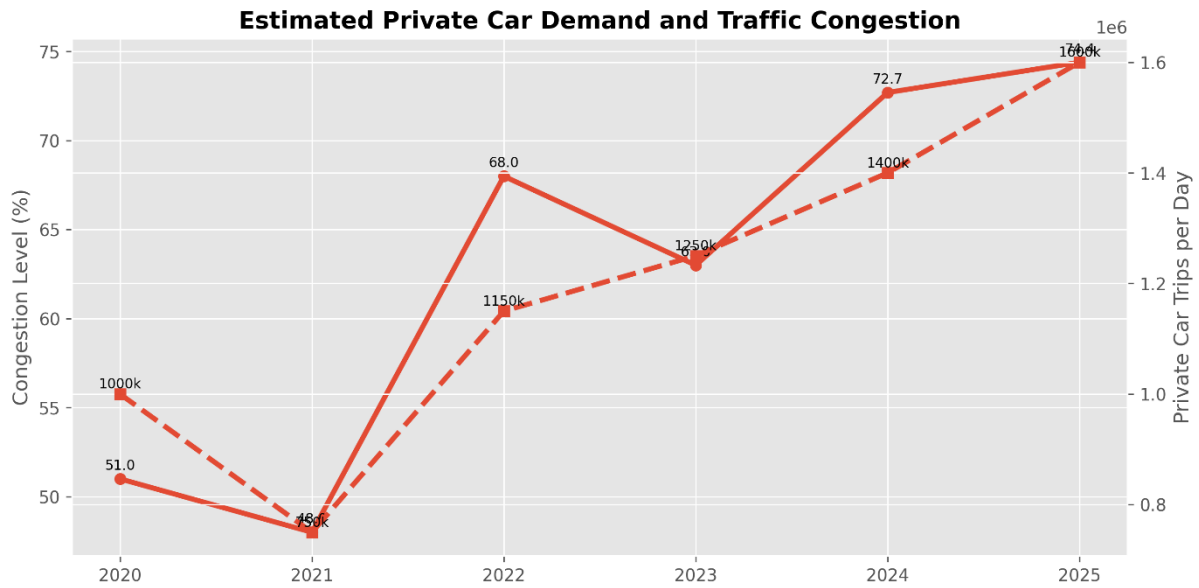
This historical traffic congestion levels of Bengaluru shows that Congestion levels generally increased across the study period, indicating that road network pressure has intensified rather than improved. Lower congestion around 2021 may reflect temporary pandemic-related mobility suppression, followed by a steady rebound as everything resumed. A rise from **51% (2020)** to **74.4% (2025)** is not a small change. It indicates a substantial worsening of congestion conditions in Bengaluru over five years. At this level, the network may behave closer to saturation during peak hours, where small increase in demand can create disproportionately larger delays.



The correlation matrix

A correlation matrix was first developed for Bengaluru to examine the indicative statistical relationships between traffic and mobility parameters and the emissions. It's necessary to understand how these variables move together. A correlation matrix helps identify which variables have strong positive relationships, which variables move inversely, which indicators are most associated with rising emissions. We have identified strong positive correlation between demand variables like private car trips, vehicle kms travelled, congestion level and the CO2 emissions. Congestion percentage showed a negative correlation with rush-hour speed. Higher congestion is associated with slower traffic movement. As years pass by the congestion level and the rush hour speed worsens. This interprets continued structural problem with the public awareness.



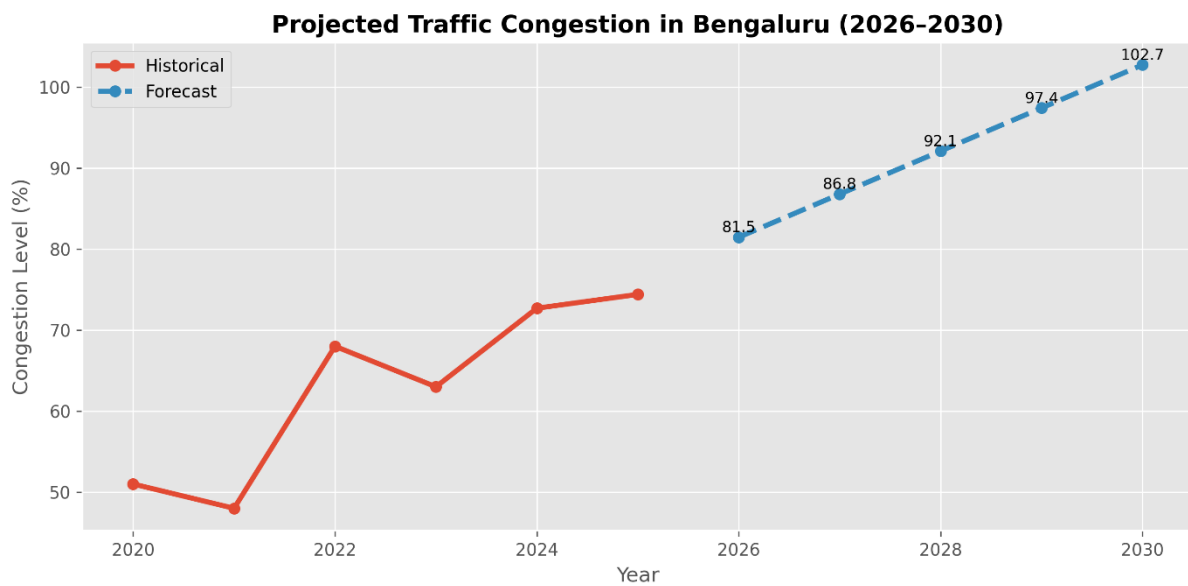


Forecasting analysis (2026-2030)

Following the exploratory correlation and trend assessment, forecasting was conducted to estimate how congestion levels may evolve from 2026 to 2030 for Bengaluru.

Forecasting Method Used - Damped Holt Exponential Smoothing

This method captures historical trend direction, gradual continuation of change, slowing growth over time (damping effect). Unlike simple linear extrapolation, it avoids unrealistic indefinite acceleration. Since the study used annual observations, exponential smoothing performs well with limited structured data with no seasonal trends. Forecasts indicated that Bengaluru is likely to experience continued increases in congestion intensity if no measures are taken w.r.t single occupant private cars. Forecasts show a very steady increase striking as much as 102.7% congestion level by 2030.



Elasticity analysis

Elasticity measures how sensitive congestion is to changes in congestion relevant private car trips.

If private-car trips change by 1%, by how much percent does congestion change?

Elasticity = %change in congestion %/ % change in congestion relevant trips

We arrived at the elasticity 0.76 which means that a 1% increase in congestion relevant car trips leads to 0.76% increase in congestion %.

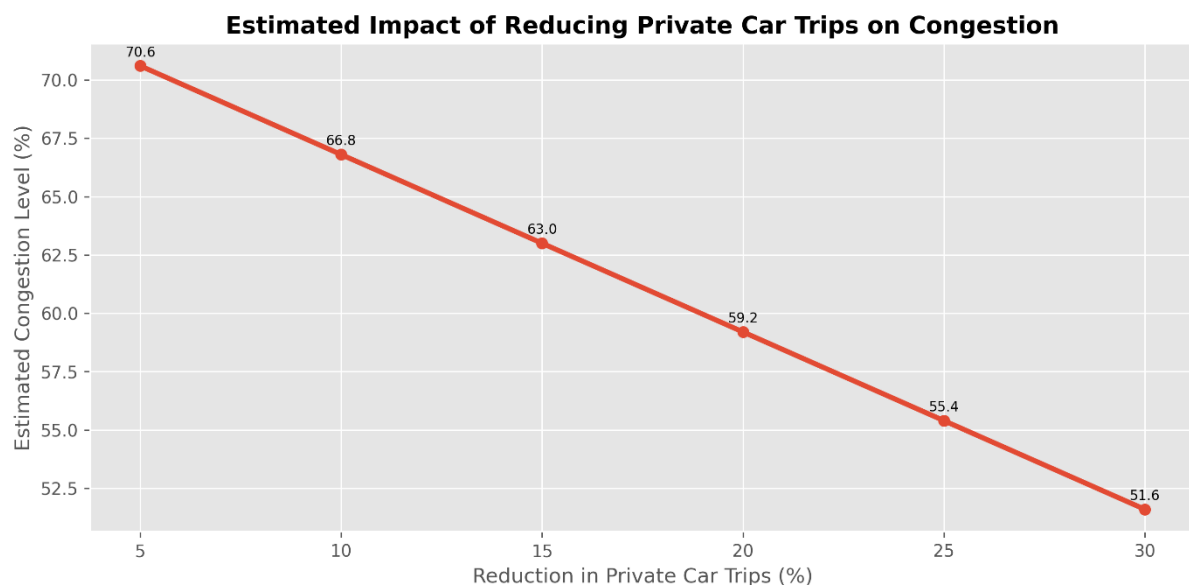
This indicates that reducing private-car demand during peak hours could generate meaningful benefits reducing congestion and improving rush hour speed.

Trip reduction Scenario modelling

To estimate how congestion levels may improve if private car trips decline under various policies. These represent realistic intervention ranges achievable through combined measures such as public transport modal shift and hybrid work adoption.

% reduction in congestion = Elasticity * % reduction in private car trips

So, if trip reduces by 5% then congestion % reduces by 3.8% ($0.76 \times 5\%$) leading to drop in congestion % to 70.6%. This means that even small reduction in private car trips leads to considerable congestion relief. Managing unnecessary private car trips can have huge impact that the new road capacity. Incremental reductions in car demand could generate proportionate improvements in Bengaluru's congestion performance.



CO₂ Savings and Revised Emissions under Trip Reduction Scenarios

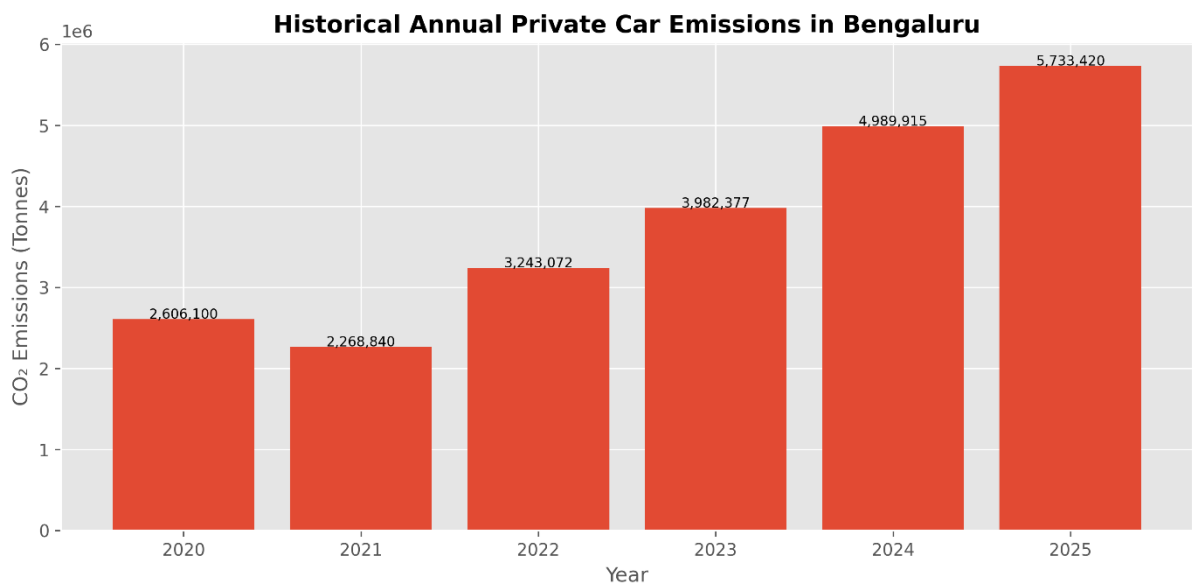
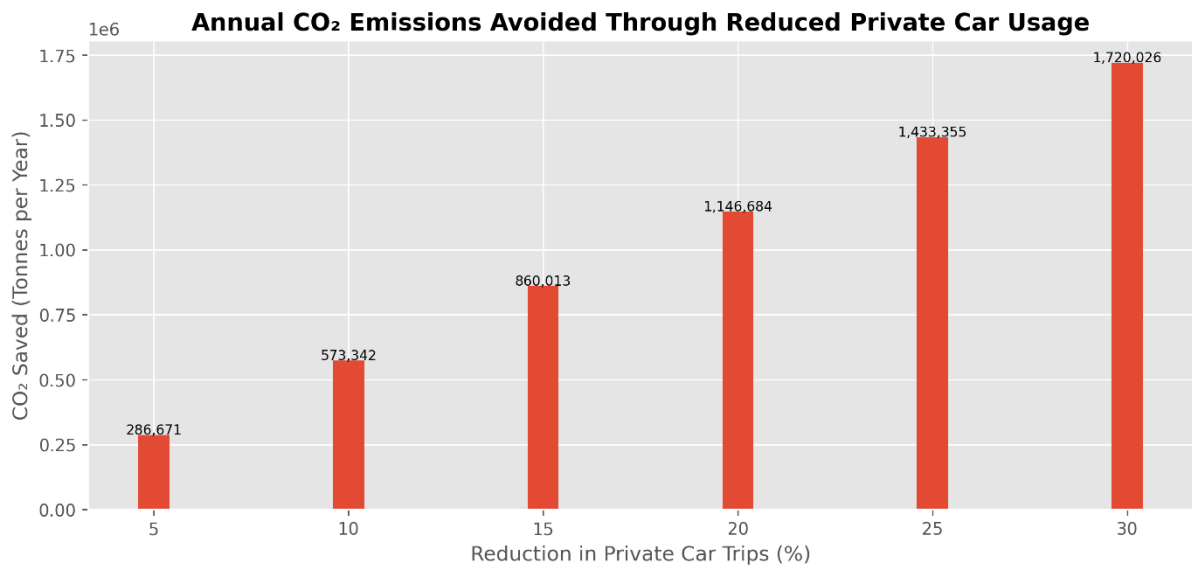
For each trip-reduction scenario, the study calculated - New CO₂ emissions (tonnes) after trip reduction, CO₂ emissions saved (tonnes) relative to the baseline scenario. It answers If Bengaluru reduces private-car trips by 5%, 10%, 15%, 20%, or 30%, how many tonnes of CO₂ could be avoided annually? This measures environmental impact. Taking 2025 as baseline, we calculate

New trips = Baseline trips * (1 – Reduction %)

New CO₂ emissions = Baseline CO₂ * (1 – Reduction %)

CO₂ saved = Baseline CO₂ – New CO₂

This analysis infers that even a modest reduction leads to significant avoidable emissions which is in line with the congestion %.



Benchmarking with other cities around the world

We have done comparative analysis of congestion, travel efficiency, emission trends of Bengaluru with global cities like Zurich, Singapore, and Amsterdam. Benchmarking is used to understand Bengaluru’s place in global context and identify and adopt practical improvement paths. A single city analysis doesn’t provide complete picture of the global context. Hence, Benchmarking helps to interpret whether the congestion levels observed in Bengaluru are unusually high or moderate or in line relative to the compared cities of the world.

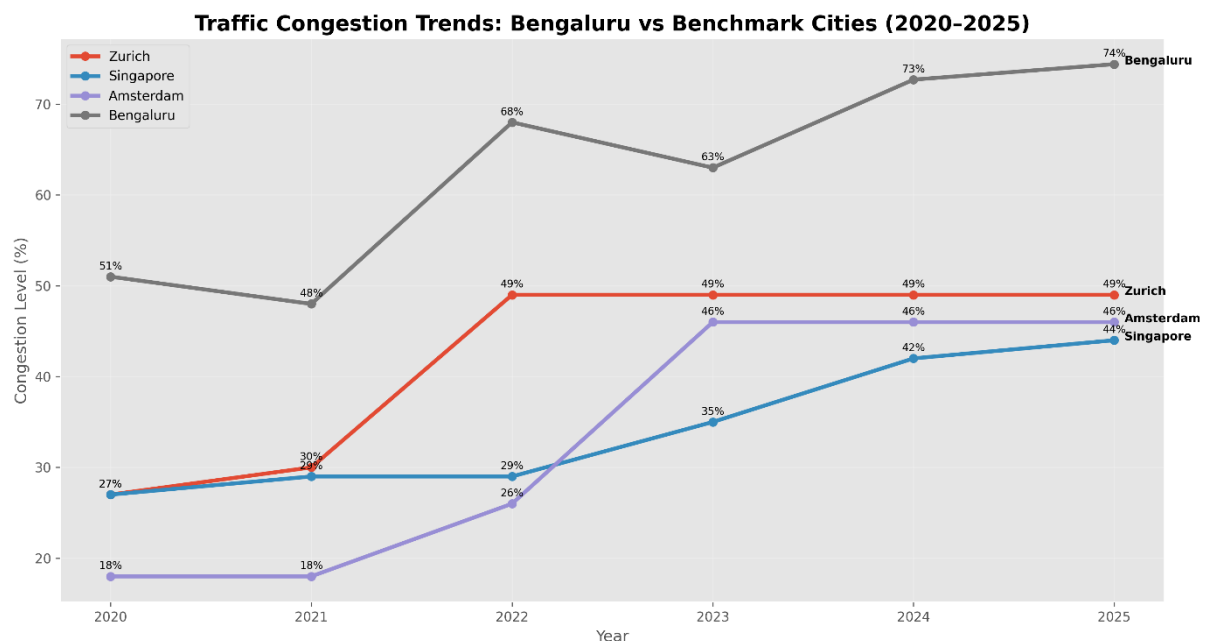
This comparative analysis helps reveal areas where Bengaluru is lagging such as Congestion intensity, travel efficiency due to single occupant cars, emission outcomes. Benchmarking allows us to study the lessons to be drawn from the cities that successfully implemented mobility strategies. While population plays a major factor, necessary actions need to be taken to ensure to tackle the problem of

congestion due to single occupant private cars wherever possible. Cross city comparisons make recommendations more credible because they provide best international mobility practices.

Zurich is selected for benchmarking as it has high quality public transport, strong planning discipline and represents a mature, high efficiency mobility model with low car dependency and sustainable practices widely adopted across the city. Despite high income levels and ability to own cars, Zurich maintains relatively efficient transport outcomes. Singapore is one of the most cited global examples of actively managing congestion through ownership controls, vehicle quota systems, integrated planning which is highly relevant for cities with road space scarcity. Amsterdam is internationally recognized for reducing car dependence through cycling networks, pedestrian priority, mixed urban planning aligning mobility with emission goals.

Congestion trend analysis

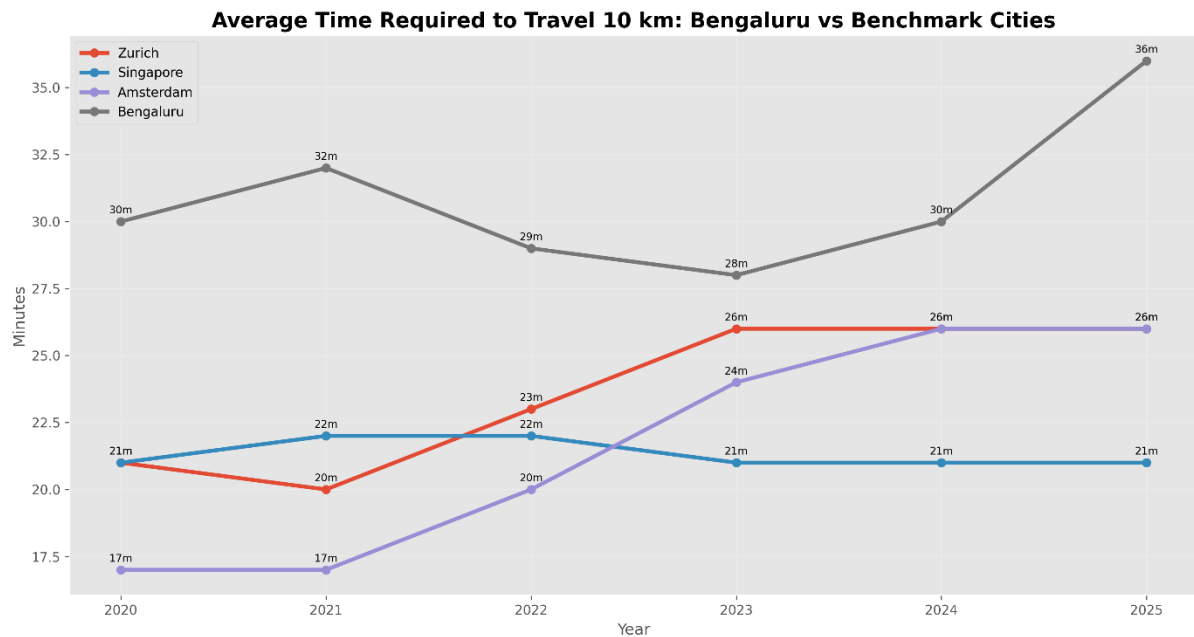
As a part of the benchmark analysis, we have compared the historical congestion trends between Bengaluru and the selected benchmark cities from 2020 till 2025. This shows the congestion trends with directions of change over time and relative ranking across cities. Benchmark cities displayed a controlled trajectory over Bengaluru. Zurich and Amsterdam have been utmost consistent for the past 2 years with much controlled mobility while Singapore shows a little sharp rise in 2024 compared to 2022. While Zurich saw a considerable rise from 30% in 2021 to 49% in 2022 which is rise in more than 50% of 2021 level. This shows similar pattern with the raising Bengaluru congestion level in 2022 relative to 2021 and Amsterdam as well. This reflects post pandemic normalization and mobility rebound.



Comparison of average time required to travel 10 km.

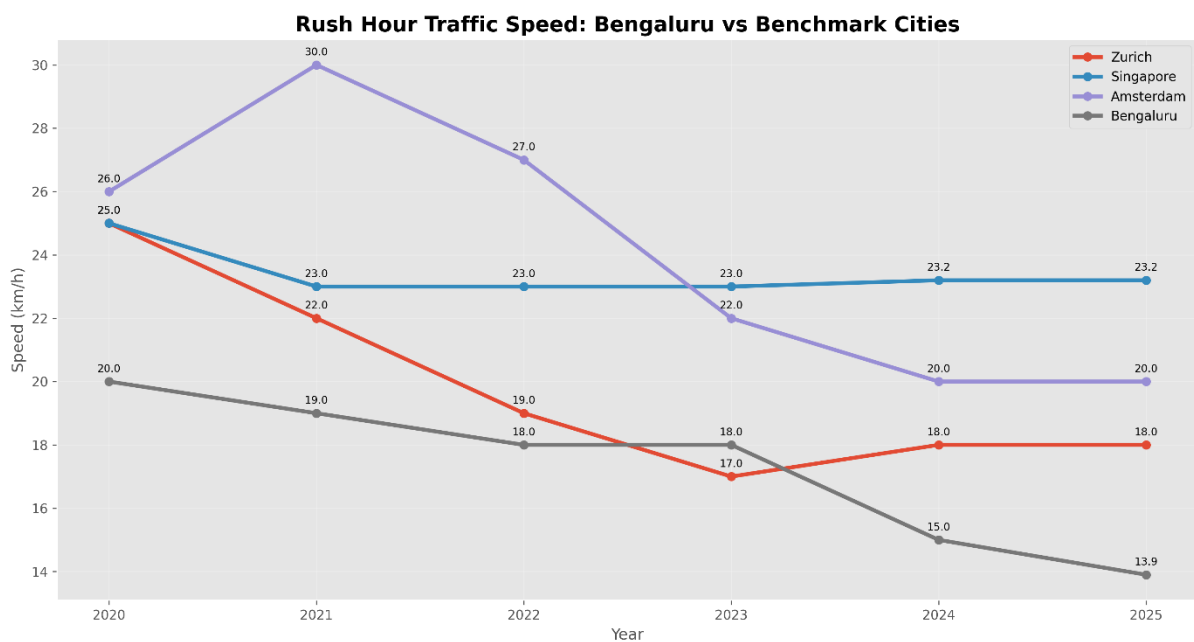
We have also compared the average time required to travel 10 kms between Bengaluru and selected benchmark cities from 2020 till 2025. While Bengaluru's average time worsens in 2025 at 36 mins per 10 km (20% in one year) there has been a controlled mobility in benchmark. This infers that the congestion level is worst affected due to the single occupant private cars leading to a bigger average

time. Singapore has been consistently at 22 mins showing strong network efficiency. Zurich and Amsterdam have increased till 2023 but stabilized later due to efficient public transport.



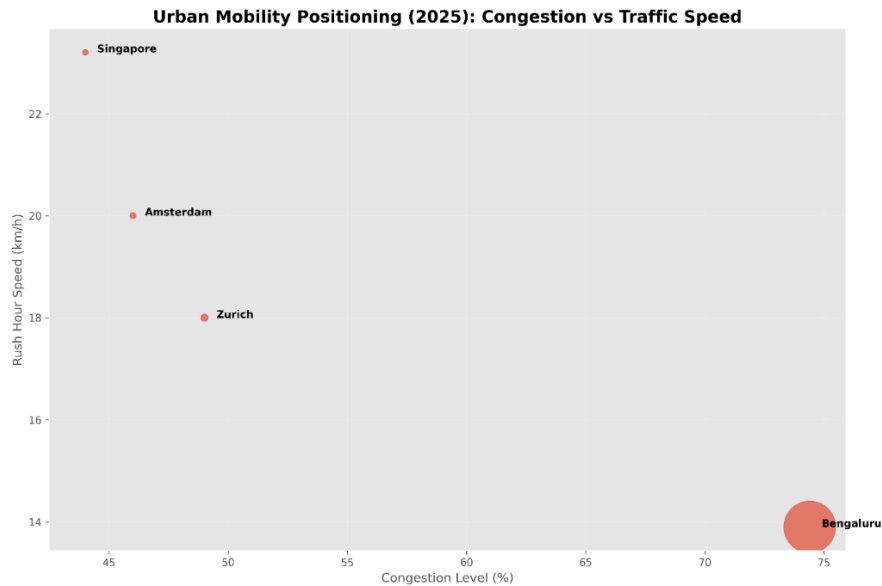
Rush hour traffic speed analysis

The rush hour traffic speed has declined from 2020 till 2025 for all the cities. But for Bengaluru it has worsened at 13.9 km/hr in 2025 which is significantly lower than all the benchmark cities. Singapore has been consistent with strong network mobility and policy framework. The other cities have more than 35% of Bengaluru rush hour speed. This demands the need for Bengaluru to take necessary actions on reducing the unnecessary congestion trips during rush hours.



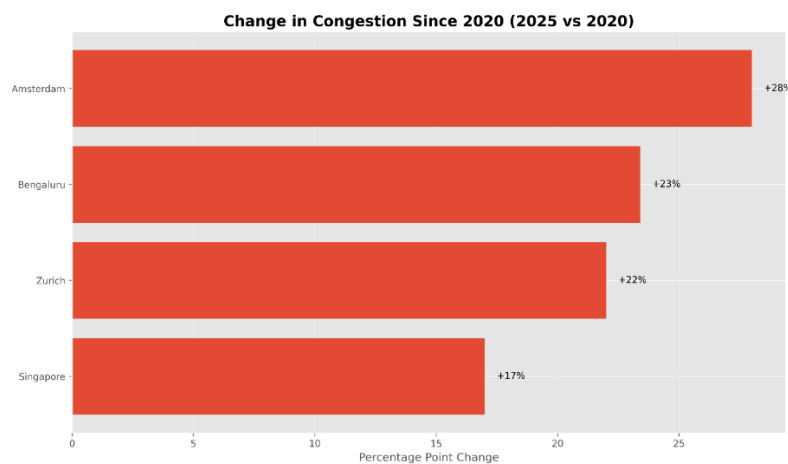
2025 Urban mobility positioning

The urban mobility positioning map comparing Bengaluru with benchmark cities using three indicators - Congestion Level, Rush Hour Speed while Bubble Size represents rank position (Top rank – Larger bubble indicating worse Traffic position). Top left shows the best position while bottom right represents worse position. Bengaluru is on the bottom right which larger bubble showing the worst position, Singapore represents a strong urban mobility on the top left with a smaller bubble.



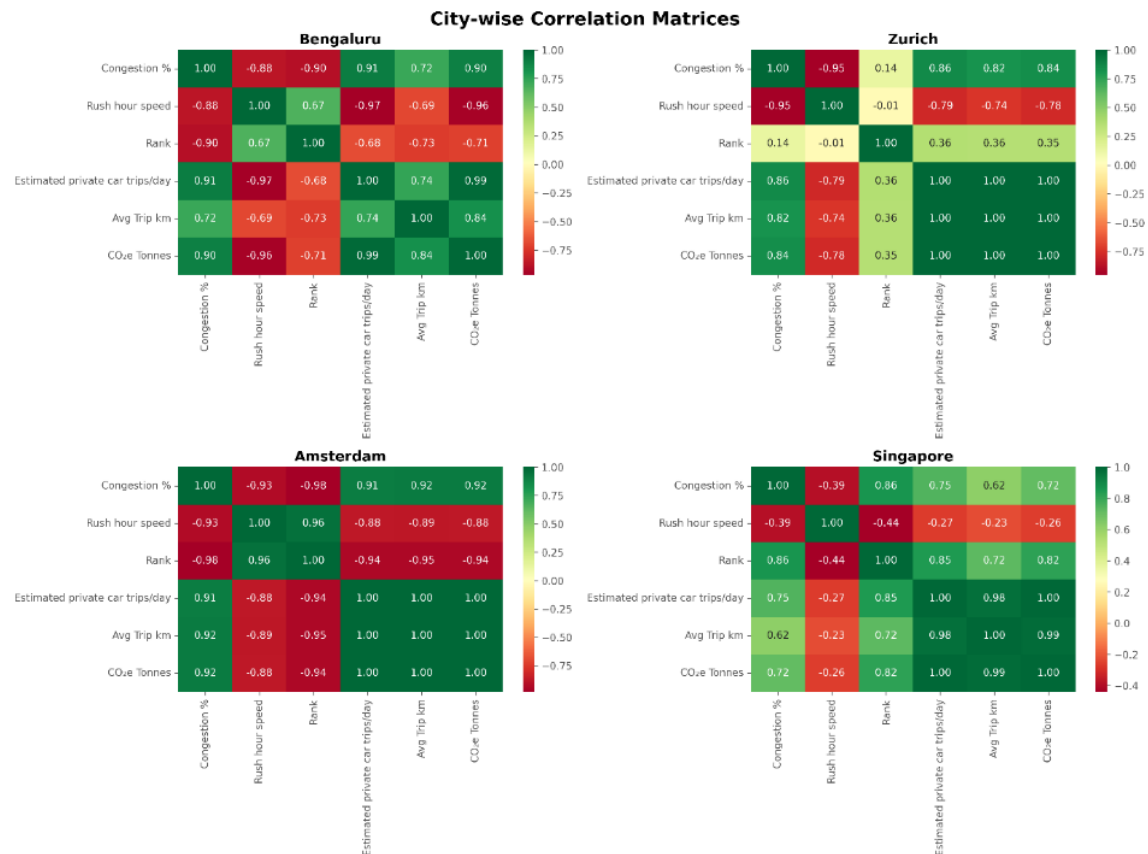
Analysis of change in congestion since 2020

The analysis shows that there has been highest raise in congestion level from 2020 for Amsterdam city(28%) especially in 2023 post pandemic and has stabilized post that. Bengaluru has the second highest change of around 23% rise in 2025 compared to 2020 which has been consistently raising.



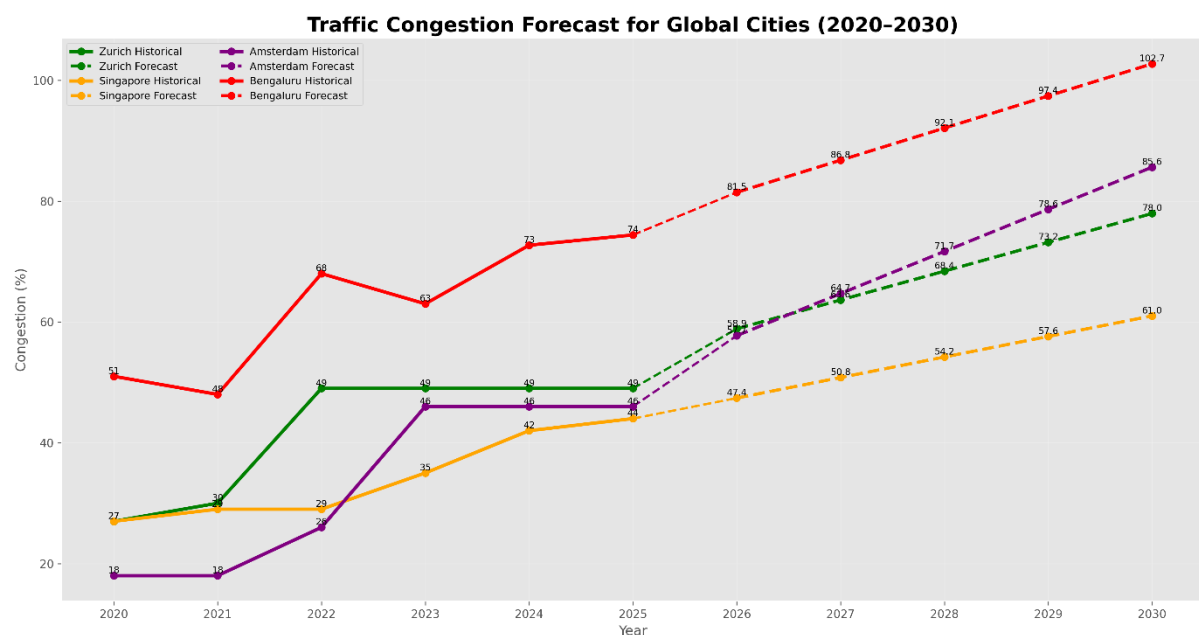
City wise correlation matrices

We have constructed city wise correlation matrices for all the 4 cities. We have identified strong positive correlation between demand variables like private car trips, vehicle kms travelled, congestion level and the CO2 emissions. Congestion percentage showed a negative correlation with rush-hour speed. Higher congestion is associated with slower traffic movement. This shows that the structural problem of private car trips per day leads to congestion level.



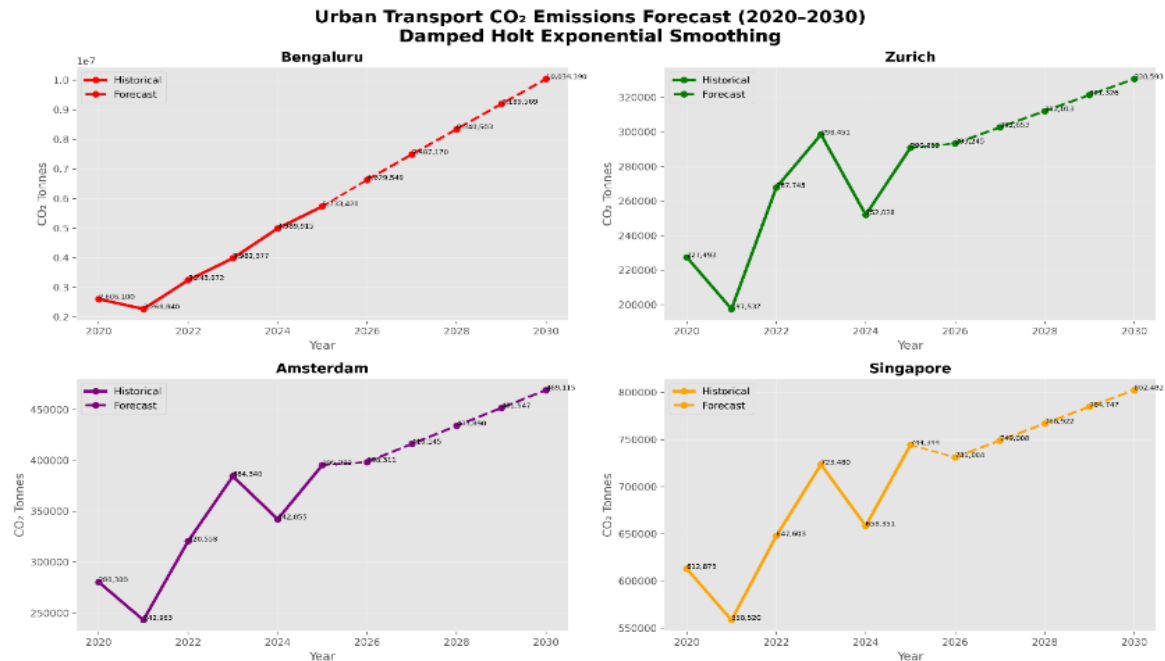
Traffic congestion forecast for global cities

From the analysis we can infer that Bengaluru has been forecasted at 102.7 % congestion level by 2030 if no disciplined effort and action is taken. We can observe in the chart that the Amsterdam congestion level is expected to rise higher than Zurich due to its past historical trend shift. Since It rose swiftly from 26% in 2022 to 46% in 2023 the forecast takes into account the behavioural patterns to project future forecasts. The chart shows that all cities scale upward pressure, but the rate and scale vary considerably.



Urban CO2 emissions forecast till 2030.

We have estimated the CO2 e forecast using Damped Holt Exponential smoothing. The chart shows that the emissions would be around 10 Mn CO2 tonnes pa due to the estimated private car trips. Singapore is expected around 0.8 Mn CO2 tonnes pa since the estimated private car trips per day are high, but congestion relevant car trips are lesser leading to better mobility yet high emissions.



Trip reduction scenario analysis

We have performed scenario modelling to estimate the reduction in CO₂ emissions with various scenarios like trip reduction, EV transition, shorter average trips as follows.

Scenario

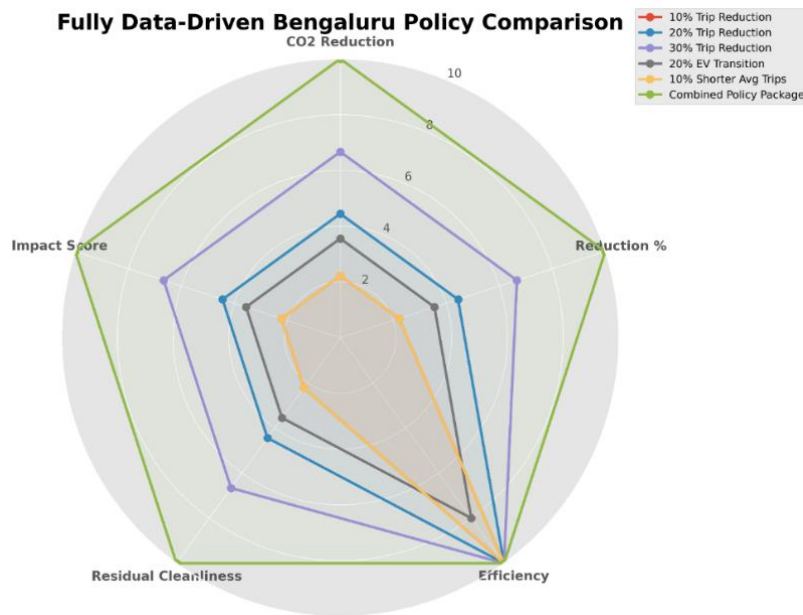
- 1 No Action Baseline
- 2 10% Trip Reduction
- 3 20% Trip Reduction
- 4 30% Trip Reduction
- 5 20% EV Transition (Lower effective emission factor)
- 6 10% Shorter Avg Trips
- 7 Combined Policy Package (Trip reduction, EV Transition, Shorter avg trips)

The radar chart compares each scenario across five normalized performance dimensions:

Axis	Meaning
CO ₂ reduction	Emissions saving potential
Reduction %	Relative reduction compared to baseline
Efficiency	Implementation effort
Residual cleanliness	Remaining emissions performance

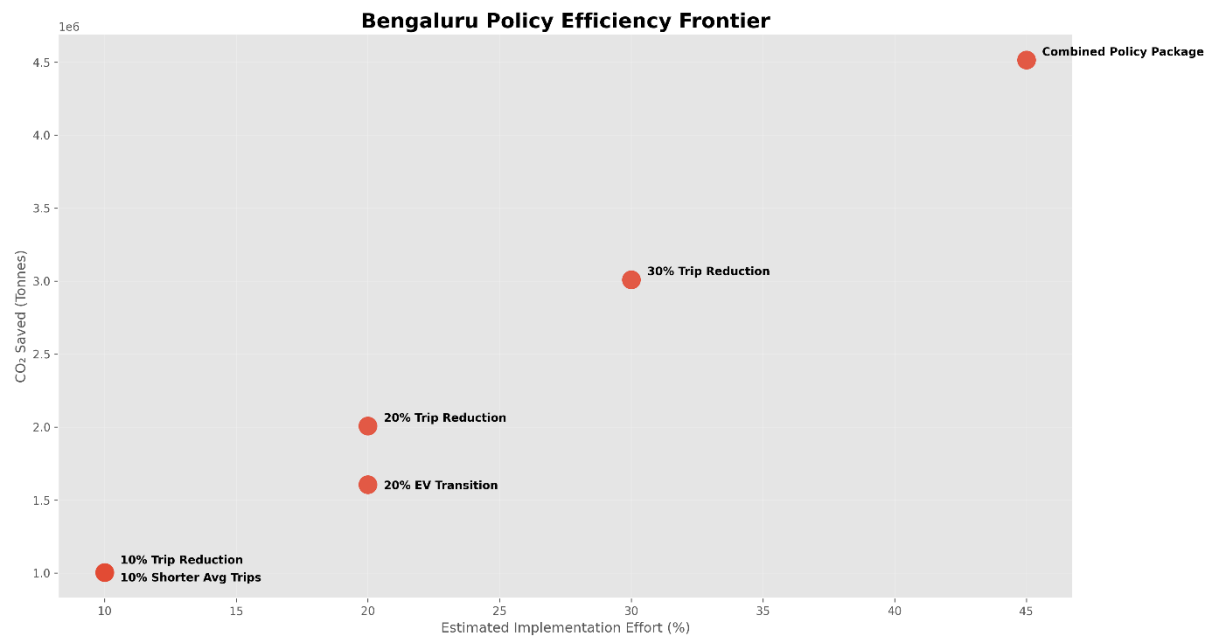
Impact score	Overall composite policy benefit
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Scenario modelling showed that isolated interventions provide moderate benefits, whereas integrated mobility packages produce substantially larger emissions reductions. It shows that 30% trip reduction saves 3 million tonnes annually. However, EV transition alone is not enough as it saves less than 20% trip reduction due to low effective emission factor.



Bengaluru policy efficiency frontier

This shows the implementation effort required to achieve reduced emissions. Combined policy package needs 45% implementation effort which could not be achieved without public awareness and the policy intervention by public authorities. The implementation effort combined different practical dimensions the behaviour change difficulty, policy complexity, cost, and the time to deploy. The % scale we have taken is a normalized relative score representing low, moderate, high, and very high effort required for each scenario. While 10% trip reduction requires lower effort, 20% trip reduction requires moderate effort, 30% reduction requires high effort with substantial reforms and combined policy package requires the highest effort simultaneously demanding actions.



Inference

The study findings showed that Bengaluru faces

Persistently high and rising congestion

Worsening Rush hour speed and 10 km travel times

High projected 2030 transport emissions

Widening gap versus benchmark cities

Strong sensitivity of congestion to single occupant private car trip volume.

This indicates that **road space is increasingly scarce**, while a large share of trips is being consumed by low-occupancy private vehicles. So, this study urges on the need to reduce the single occupant private car trips.

Solution framework

Peak hour congestion pricing for core corridors

Introduce dynamic pricing for single occupant private cars entering the most congested corridors during peak periods such as key IT commuting belts and busy zones. It directly reduces the unnecessary peak hour demand leading to increase in rush hour speed and reduction in average per 10 km time.

Employer led mobility mandate for Tech parks.

Requires the employers to submit annual commute reduction plans using hybrid work, flexible timings, corporate shuttles since Bengaluru congestion is strongly linked to concentrated commute peaks with Single occupant private cars of the employees.

2-day hybrid work mobility impact, wherever possible can materially reduce congestion level based on the trip reduction scenario analysis performed.

Public transport mobility integration

Integrate metro stations and buses with all the areas, shuttles, shared autos, walking corridors since many people drive because the last mile is broken. The connectivity across all areas with sufficient number of buses and metros can reduce the single occupant private cars significantly.

Real time AI Traffic command centre

Using AI for corridor congestion prediction to identify the routes causing congestion and performing the root cause analysis of the private car trips and taking necessary actions to minimize the single occupant private car trips.

Mobility carbon budget for the city

Set an annual transport CO₂ budget for Bengaluru and align policies to remain within it. To monitor the single occupant private car trips, congestion, emissions. If threshold breaches then take necessary measures like increase parking prices, provide incentives.

Personal congestion footprint dashboard

An App based estimation with Automatic detection of mobility impact of each private car with single occupant, solo car km, congestion contribution, CO₂ emitted, alternatives available with policies capping certain metrics etc. This can lead to taking necessary actions against those exceeding the benchmarked metrics. This would require considerable implementation effort, but if introduced can have a big impact.

Introduce best practices of global countries.

For example, Singapore implemented vehicle ownership restraint making cars expensive and tight regulations. Congestion pricing, Reliable public transport, and integrated urban planning has made Singapore one of the best cities with controlled traffic despite being an economic corridor. Zurich, having world class public transport system with seamless ticketing. Amsterdam was designed for the people first and the car convenience is limited with high parking prices. It also promotes cycling infra and walking corridors for shorter trips. These best practices can be implemented in phases to make Bengaluru a less congested city.

Impact

If Private Car trips Fall:

Less Cars on Road



Higher Traffic Flow (Less congestion)



Shorter Commute Time



Lower Fuel Use



Lower Emissions

Why Bengaluru Is Ideal for Reform

Bengaluru has:

- Severe congestion
- Large knowledge-worker base
- Growing metro network
- Strong demand for productivity gains
- National visibility as a tech city

This makes it suitable for policy innovation.

Conclusion

Bengaluru's transport problem is not only about roads or fuel type. It is also about how urban travel demand is structured. A city with too many routine private car trips will remain congested even with cleaner vehicles.

The long-term solution is to reduce avoidable private car dependence while improving alternatives. Hybrid work, better public transport, smarter parking, and transit-focused planning can make Bengaluru more efficient and liveable.

Bengaluru does not only need cleaner cars. It needs fewer unnecessary car trips.

References (Public Sources)

1. TomTom Traffic Index (2025)
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3. Directorate of Urban Land Transport Karnataka
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